SECTION 3: NETWORKING BASICS

Task 3.1: Network Concepts

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1. OSI Model

The OSI (Open Systems Interconnection) Model is a conceptual framework used to understand how communication happens in a network. It divides the communication process into 7 layers, each with a specific function.

<u>The 7 layers of the OSI model are:</u>

1. Physical Layer – Responsible for transmitting raw data bits over a physical medium such as cables or signals.
   Example: Ethernet cables
2. Data Link Layer – Handles physical addressing (MAC address) and error detection.
   Example: Ethernet
3. Network Layer – Responsible for logical addressing and routing of data packets.
   Example: IP (Internet Protocol)
4. Transport Layer – Ensures reliable data transfer, error correction, and flow control.
   Example: TCP, UDP
5. Session Layer – Manages sessions and connections between devices.
   Example: NetBIOS
6. Presentation Layer – Handles data formatting, encryption, and compression.
   Example: SSL/TLS
7. Application Layer – Provides services directly to users such as web browsing and email.
   Example: HTTP, FTP, SMTP

TCP (Transmission Control Protocol) operates at the Transport Layer (Layer 4). It ensures reliable and ordered delivery of data.

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2. IP Addressing

An IP address is a unique identifier assigned to devices in a network.

Difference between Public IP and Private IP:

Public IP:

- Accessible over the internet
- Globally unique
- Assigned by Internet Service Provider (ISP)
- Less secure because it is exposed

Private IP:

- Used within local networks
- Not accessible directly from the internet
- Assigned by router
- More secure

Private IP ranges are:

Class A: 10.0.0.0 to 10.255.255.255
Class B: 172.16.0.0 to 172.31.255.255
Class C: 192.168.0.0 to 192.168.255.255

192.168.1.100 is a Private IP address because it falls within the Class C private range.

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3. Protocols

Matching ports to services:

Port 22 – SSH (Secure Shell) – Used for secure remote login
Port 80 – HTTP – Used for web browsing (non-secure)
Port 443 – HTTPS – Used for secure web browsing
Port 3306 – MySQL – Used for database services

Difference between TCP and UDP:

TCP:
- Connection-oriented
- Reliable communication
- Error checking and retransmission
- Slower but accurate

UDP:
- Connectionless
- Faster communication
- No guarantee of delivery
- Used in streaming and gaming

NAT (Network Address Translation):

NAT is a technique used by routers to convert private IP addresses into a public IP address before sending data to the internet.

Why NAT is needed:

- Saves public IP addresses
- Provides security by hiding internal IP addresses
- Allows multiple devices to share a single public IP

Task 3.2: Practical Network Analysis

Commands used:

ifconfig
route -n
netstat -tuln
ping google.com
traceroute google.com

Explanation of results:

IP Address:
The IP address identifies the system in the network. In most home networks, it is a private IP like 192.168.x.x.

Default Gateway:
The default gateway is the router's IP address. It is used to send data outside the local network. Example: 192.168.1.1

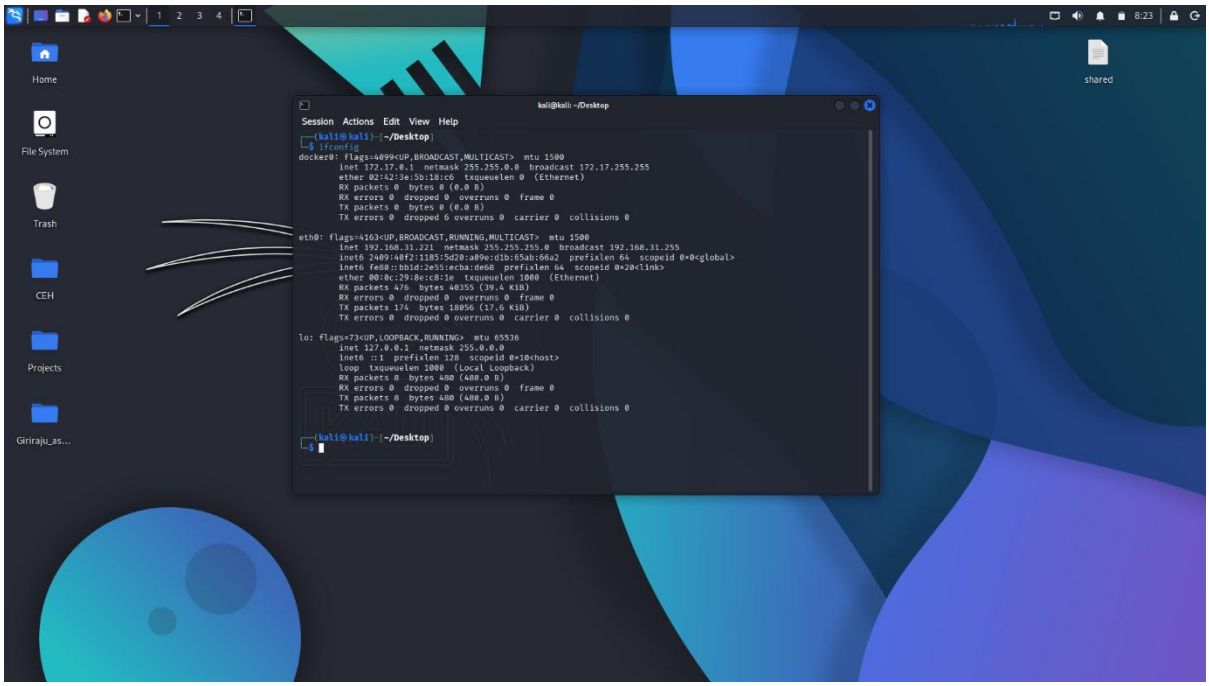
DNS Servers:
DNS servers convert domain names into IP addresses. Example: 8.8.8.8 (Google DNS)

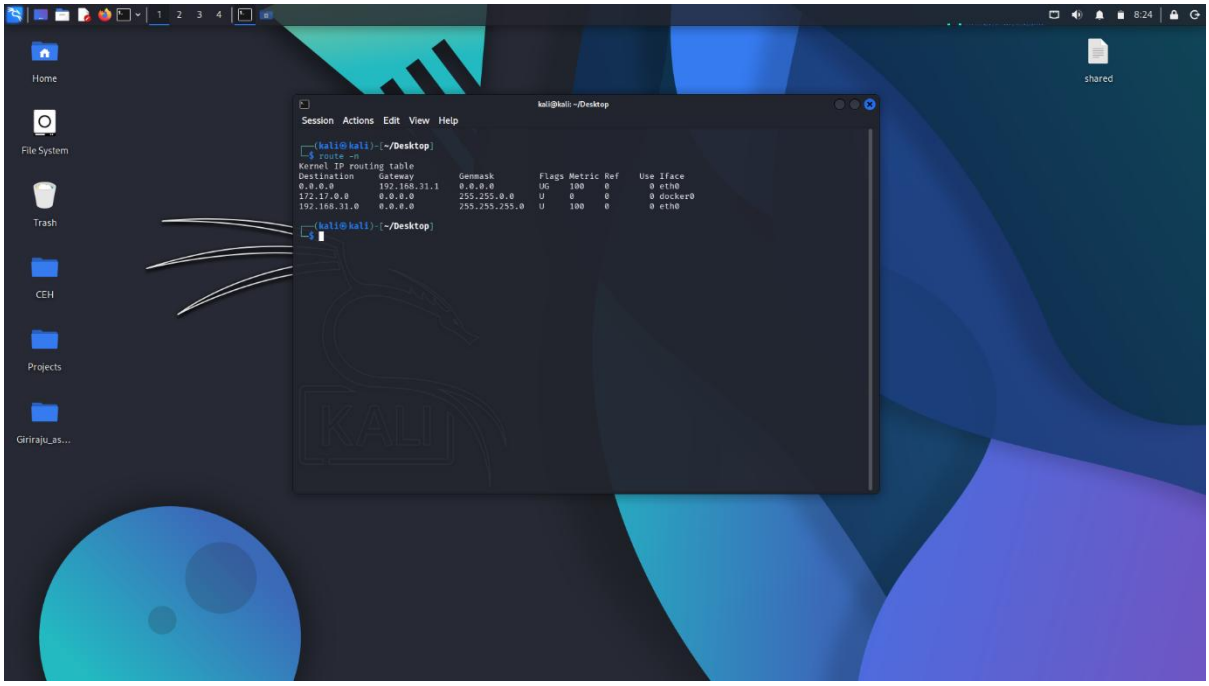
Ping Result:
Ping is used to test connectivity. It shows response time and packet loss. Low time and 0% loss indicate a good connection.

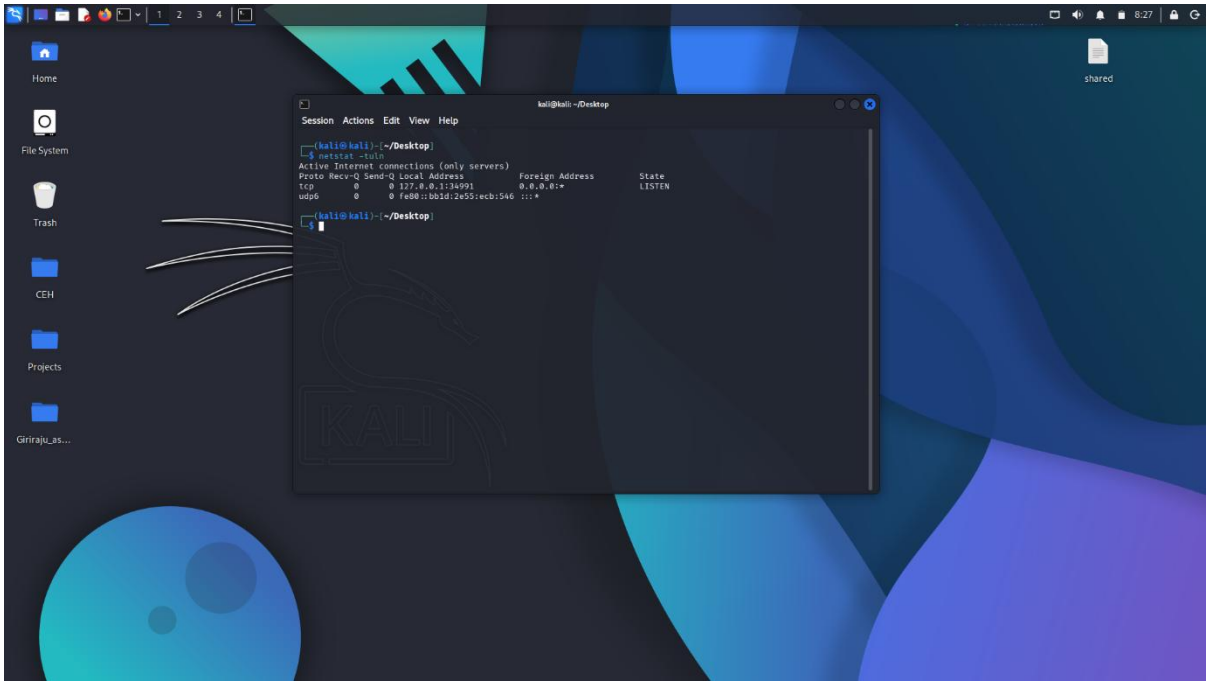
Traceroute:
Traceroute shows the path taken by data to reach a destination. Each step is called a hop. Usually, it takes around 8–15 hops to reach google.com.

Netstat:
Netstat displays active network connections and open ports. It helps in identifying running services.

Routing Table:
The routing table shows how data is directed through the network. It includes destination networks and gateway information.


8:23
Home
File System
Trash
CEH
Projects
Giriraju_as...
shared
kali@kali: ~/Desktop
Session Actions Edit View Help
┌──(kali㉿kali)-[~/Desktop]
└─$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
ether 02:42:3e:5b:18:c6 txqueuelen 0 (Ethernet)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 6 overruns 0 carrier 0 collisions 0
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 192.168.31.221 netmask 255.255.255.0 broadcast 192.168.31.255
inet6 2409:40f2:1185:5d20:a09e:d1b:65ab:66a2 prefixlen 64 scopeid 0x0<global>
inet6 fe80::bb1d:2e55:ecba:de68 prefixlen 64 scopeid 0x20<link>
ether 00:0c:29:8e:c8:1e txqueuelen 1000 (Ethernet)
RX packets 476 bytes 40355 (39.4 KiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 174 bytes 18056 (17.6 KiB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0x10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 8 bytes 480 (480.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 8 bytes 480 (480.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
┌──(kali㉿kali)-[~/Desktop]
└─$


8:24
Home
File System
Trash
CEH
Projects
Giriraju_as...
shared
kali@kali: ~/Desktop
Session Actions Edit View Help
┌──(kali㉿kali)-[~/Desktop]
└─$ route -n
Kernel IP routing table
Destination Gateway Genmask Flags Metric Ref Use Iface
0.0.0.0 192.168.31.1 0.0.0.0 UG 100 0 0 eth0
172.17.0.0 0.0.0.0 255.255.0.0 U 0 0 0 docker0
192.168.31.0 0.0.0.0 255.255.255.0 U 100 0 0 eth0
┌──(kali㉿kali)-[~/Desktop]
└─$


Home
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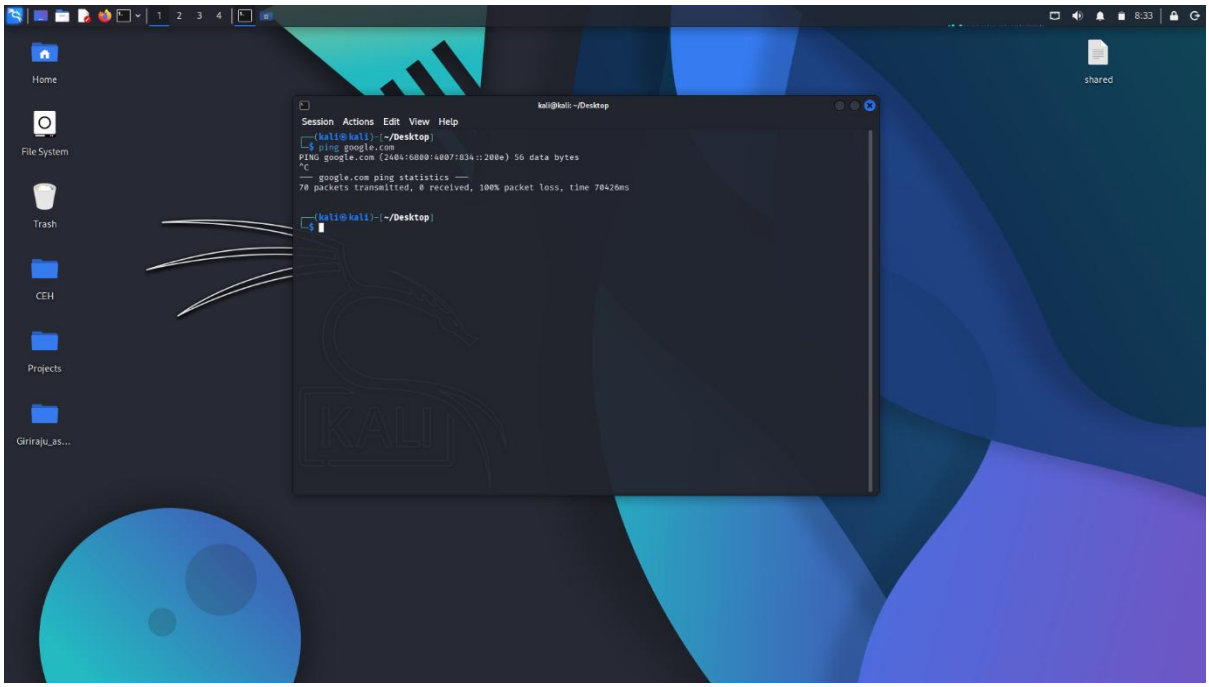
shared
kali@kali: ~/Desktop
Session Actions Edit View Help
┌──(kali㉿kali)-[~/Desktop]
└─$ netstat -tuln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address Foreign Address State
tcp 0 0 127.0.0.1:34991 0.0.0.0:* LISTEN
udp6 0 0 fe80::bb1d:2e55:ecb:546 :::*
┌──(kali㉿kali)-[~/Desktop]
└─$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Giriraju_as...
shared
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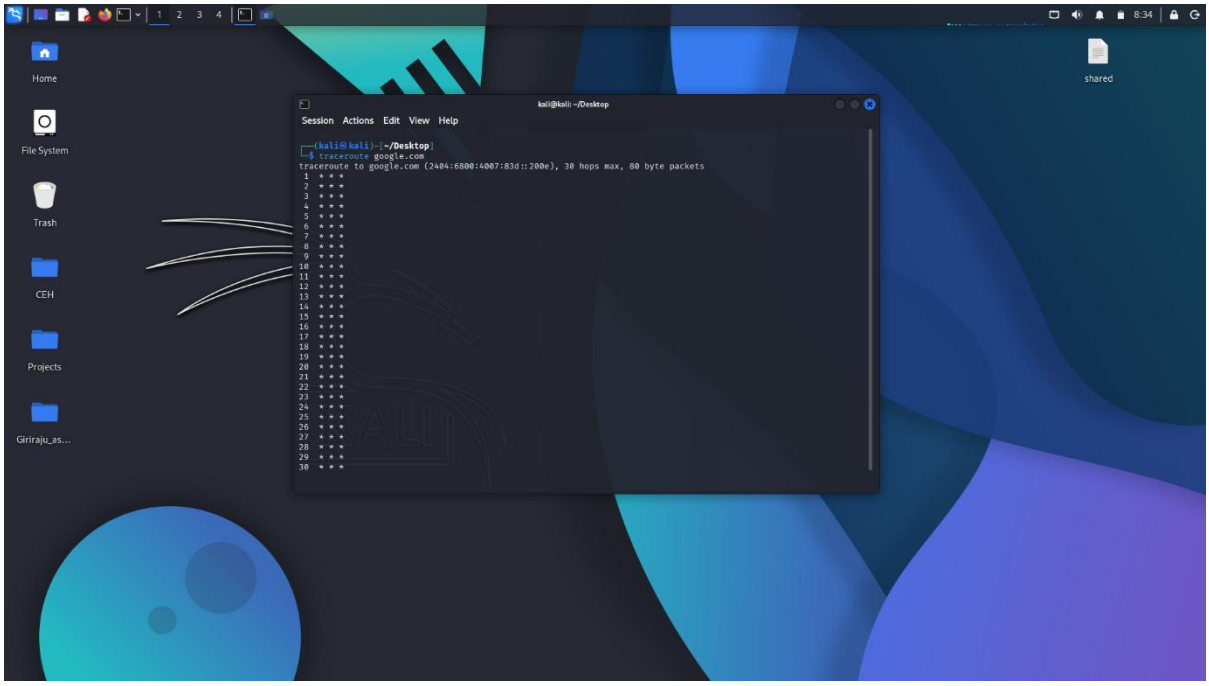
Session Actions Edit View Help
┌──(kali㉿kali)-[~/Desktop]
└─$ ping google.com
PING google.com (2404:6800:4007:834::200e) 56 data bytes
^C
--- google.com ping statistics ---
70 packets transmitted, 0 received, 100% packet loss, time 70426ms
┌──(kali㉿kali)-[~/Desktop]
└─$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Giriraju_as...
shared
kali@kali: ~/Desktop
Session Actions Edit View Help
┌──(kali㉿kali)-[~/Desktop]
└─$ traceroute google.com
traceroute to google.com (2404:6800:4007:83d::200e), 30 hops max, 80 byte packets
1 * * *
2 * * *
3 * * *
4 * * *
5 * * *
6 * * *
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